

MATH 5610/6610: Advanced Numerical Analysis**Homework 1**

Due: Friday, September 25, 2026

Instructions. Show all work. For the computing problem, submit your code and its output together with a short discussion. You may use MATLAB or Python.

Problem 1. Let

$$A = \begin{bmatrix} 1 & 2 & 2 \\ 4 & 4 & 2 \\ 4 & 6 & 4 \end{bmatrix}, \quad b = \begin{bmatrix} 3 \\ 6 \\ 10 \end{bmatrix}.$$

Compute the factorization $A = LU$ by Gaussian elimination without pivoting, recording the multipliers. Then solve $Ax = b$ by one forward substitution ($Ly = b$) and one back substitution ($Ux = y$). Verify your solution by computing Ax .

Problem 2. Write the elimination matrices as $M_k = I - c^{(k)}e_k^T$, where the vector $c^{(k)}$ contains the multipliers of step k and satisfies $(c^{(k)})_j = 0$ for $j \leq k$. Gaussian elimination gives $M_{n-1} \cdots M_2 M_1 A = U$, hence $A = LU$ with $L = M_1^{-1} M_2^{-1} \cdots M_{n-1}^{-1}$. Prove that

$$L = I + \sum_{k=1}^{n-1} c^{(k)} e_k^T,$$

that is, the multipliers drop into place with no interaction between the factors. (Hint: first show $e_j^T c^{(k)} = 0$ for $j \leq k$, then expand the product by induction on the number of factors.)

Problem 3. (Forsythe's example.) Let

$$A = \begin{bmatrix} 10^{-4} & 1 \\ 1 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

- Solve $Ax = b$ exactly (keep fractions or many digits).
- Solve it by LU factorization *without* pivoting, carrying out every arithmetic operation in 3-significant-digit decimal rounding.
- Repeat (b) *with* partial pivoting.
- Compare the three answers and explain, in terms of the sizes of the intermediate quantities, why (b) fails and (c) succeeds.

Problem 4. Suppose Gaussian elimination without pivoting runs to completion on $A \in \mathbb{R}^{n \times n}$ with all pivots $u_{11}, \dots, u_{nn}$ nonzero, so $A = LU$ with L unit lower triangular. Let A_k denote the leading principal $k \times k$ submatrix of A . Prove that

$$\det A_k = u_{11} u_{22} \cdots u_{kk}, \quad k = 1, \dots, n,$$

and conclude that all leading principal minors of A are nonzero. (Hint: partition L and U into blocks and show $A_k = L_k U_k$.)

Problem 5. (Computing.) Implement LU factorization with partial pivoting from scratch (do not call `lu`), returning P, L, U .

- (a) Test your code on a random 200×200 matrix with standard normal entries: report $\max_{ij} |(PA - LU)_{ij}|$ and $\max_{jk} |l_{jk}|$.
- (b) Report the growth factor $\rho = \max_{jk} |u_{jk}| / \max_{jk} |a_{jk}|$ for ten random matrices. How do your values compare with the worst-case bound 2^{n-1} ?
- (c) Solve $Ax = b$ with your factors for a b constructed so that the exact solution is $x = (1, \dots, 1)^T$, and report $\|x_{\text{computed}} - x\|_{\infty}$.